A MORPHISM INDUCING ISOMORPHISMS OF AUTOMORPHISM GROUPS INJECTS THE POINTS OF A CARTESIAN SQUARE OF STACKS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The description of morphisms into a 2-fibre product of stacks as triples consisting of two objects and a 2-morphism. Used at the start of the counted proof (source lines 10147-10150). Turns the comparison map of the statement into a statement about triples.
- The identification of the automorphism group algebraic spaces of two isomorphic points under the assumption of the statement. Referred to in the statement and used at the end of the counted proof (source lines 10152-10155). Allows the difference of two 2-morphisms to be lifted to an automorphism of the source point.

Source reference. Original remark remark-identify-automorphism-groups: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 7447–7447. Navigation pointer only: the counted text refers to the Stacks item remark-identify-automorphism-groups, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let*

$$\begin{array}{ccc} \mathcal{X}' & \longrightarrow & \mathcal{X} \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ \mathcal{Y}' & \xrightarrow{g} & \mathcal{Y} \end{array}$$

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be a cartesian diagram of algebraic stacks. If f induces an isomorphism between automorphism groups at points (Remark remark-identify-automorphism-groups), then

$$\mathrm{Mor}(\mathrm{Spec}(k), \mathcal{X}') \longrightarrow \mathrm{Mor}(\mathrm{Spec}(k), \mathcal{Y}') \times \mathrm{Mor}(\mathrm{Spec}(k), \mathcal{X})$$

is injective on isomorphism classes for any field k.

Proof. We have to show that given (y', x) there is at most one x' mapping to it. By our construction of 2-fibre products, a morphism x' is given by a triple (x, y', α) where $\alpha : g \circ y' \to f \circ x$ is a 2-morphism. Now, suppose we have a second such triple (x, y', β). Then α and β differ by a k-valued point ϵ of the automorphism group algebraic space $G_{f(x)}$. Since f induces an isomorphism $G_x \to G_{f(x)}$ by assumption, this means we can lift ϵ to a k-valued point γ of G_x. Then $(\gamma, \mathrm{id}) : (x, y', \alpha) \to (x, y', \beta)$ is an isomorphism as desired. □